

Email Inbox Lead Manager (n8n AI Agent)

Title: Autonomous Email Triaging & Lead Scoring Agent

Role: Automation Architect

Tech Stack: n8n, OpenAI (GPT-4), Gmail API / IMAP, Python, HubSpot/Notion API

Timeline: 2025

Executive Summary

"Turning the inbox from a notification stream into a structured database."

For businesses, the "Contact Us" email is a bottleneck. High-value leads get buried under spam and support tickets.

This project is an **Autonomous SDR (Sales Development Representative)** agent. It monitors incoming emails 24/7, understands the intent, extracts customer details, enriches the data, and syncs it to a CRM—all before a human even opens the inbox.

Core Technical Challenge: "Unstructured to Structured"

Emails are messy, unstructured text blocks. CRMs (like HubSpot or Salesforce) require rigid, structured data (First Name, Last Name, Company, Intent, Budget).

- **My Solution:** I engineered an **LLM Extraction Node** using strict System Prompts to parse free-text emails into a standardized JSON schema, effectively treating natural language as a database query.

System Architecture

The workflow follows a **Watch → Analyze → Act** pattern:

1. The Watchtower (Trigger Layer)

- **Protocol:** IMAP Idle / Gmail API Webhooks.
- **Function:** Triggers instantly upon receipt of a new email (Real-time, not polling).
- **Filter:** Regex pre-filtering to immediately discard obvious spam or automated newsletters, saving API costs on LLM calls.

2. The Analyst (Classification & Scoring)

- **Model:** GPT-4o Mini (Cost-optimized).
- **Logic:** Analyzes the email body to classify intent:
 - Lead (Potential Sale)
 - Support (Customer Issue)
 - Spam/Vendor (Cold Pitch)
- **Lead Scoring:** Assigns a "Priority Score" (1-10) based on keywords like "pricing," "urgent," or "enterprise."

3. The Extractor (Entity Recognition)

- **Function:** If classified as a Lead, the agent extracts key entities:
 - Contact Name
 - Company Domain
 - Phone Number
 - Pain Points (Summarized)
- **Normalization:** Python scripts standardize phone numbers to E.164 format and capitalize names for CRM consistency.

4. The Router (Action Layer)

- **CRM Sync:** Uses HTTP Request nodes to upsert (Update/Insert) the contact into HubSpot/Notion.
- **Alerting:** If Score > 8, sends a High-Priority notification to Slack/Teams with a summary and a "Draft Reply" generated by the AI.
- **Auto-Response:** For low-priority inquiries, sends a context-aware acknowledgement email.

Tech Stack Details

- **Orchestration:** n8n (Webhook-based).
- **AI Model:** OpenAI API (optimized for JSON Mode).
- **Integration:** Gmail API, Slack Webhooks, CRM REST APIs.
- **Scripting:** JavaScript/Python for data cleaning within n8n nodes.

Strategic Impact

1. **Zero Lead Leakage:** Ensures 100% of high-value leads are captured and logged immediately.
2. **Response Velocity:** Reduces "Time to First Response" from hours to seconds.
3. **Data Integrity:** Eliminates manual data entry errors, ensuring the CRM is always populated with clean, standardized data.